Devops team -sw1-s1 #5

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Title project: convert long link to short link

Project Planning and Initial Design Document: Shortify

The Problem and Solution .1.1

The Problem

- **URL Management:** Long, complex, and unwieldy URLs are a barrier to effective sharing on social media, in emails, or on .printed materials. They consume space, look unprofessional, and make tracking campaign performance difficult
- **Observability Gap:** When deploying any web service, a lack of insight into its real-time performance (user count, .request latency, error rates, stability) poses a significant risk to service quality, stability, and customer satisfaction

The Solution: Shortify (The Monitored URL Shortener)

- **Service:** Build a reliable, simple URL Shortener web service capable of handling short code creation and fast URL .redirection
- **Platform:** Implement a modern DevOps stack by **containerizing** the entire application and its monitoring infrastructure .using **Docker** and **Docker Compose**
- **Monitoring:** Instrument the service with **Prometheus** for metric collection and **Grafana** for building an actionable, real- .time dashboard to visualize health, usage patterns, and performance bottlenecks

Stakeholder Analysis .1.2

Stakeholder	Primary Interest	Influence/ Impact	Engagement Strategy
Developer / Engineer	**Clean code**, easy deployment (Docker), .and modern technology selection	High	Provide a local, self-contained, repeatable, and full- .stack environment (`docker-compose up`)
Operations / Product Manager	**Service stability**(Uptime), application .performance, KPIs, and alert status	High	Deliver a clear **Grafana Dashboard**showing real-time .usage, error rates, and latency percentiles
End User / Customer	**Speed**, reliability, and accuracy of the URL .shortening and redirection process	Medium	Ensure low request latency and robust error handling .(e.g., proper 404s for failed lookups)

Service and Database Design .1.3

Service API Endpoints (The Webservice Contract)

:The service will be built using a micro-framework (e.g., Python Flask) and will expose three primary API endpoints

Output	Input	Description	Endpoint	Method
JSON body: `{"short_code": "abc123"}`	JSON body: `{"url": "long_url_here"}`	Creates a new short code for a long URL .and stores the mapping	`shorten/`	**POST**
HTTP 302 Redirect	URL Path Variable (`short_code`)	Looks up the original URL and performs an .HTTP Redirect	`short_c>/ <ode`	**GET**
Plain text Prometheus metrics	None	Exposes custom metrics in Prometheus .format	`metrics/`	**GET**

Database Schema (SQLite)

Notes	Description	Data Type	Field
.Auto-incrementing	.Primary key	INTEGER	**id**
.**Unique Index**for fast lookup	.The unique short identifier	TEXT	**short_code**
.The target for redirection	.The original destination URL	TEXT	**long_url**
.For tracking and history	.Timestamp of creation	DATETIME	**created_at**

Key Observability Metrics .1.4

:These metrics will be instrumented in the web service and scraped by Prometheus

Week Implemented	Purpose	Metric Name / Type
Week 2	.Tracks total number of shortened URLs	Counter: `url_creation_total`
Week 2	.Tracks total successful redirects	Counter: `redirect_success_total`
Week 2	.Tracks total failed lookups (404s)	Counter: `redirect_failure_total`
Week 2	.Measures latency for POST and GET requests	Histogram: `request_latency_seconds`

UI/UX Design and Visualization (Grafana Dashboard) .1.5

A. Webservice Frontend (User UX)

- .**Design:** Minimalist, single-page HTML interface
- .**Functionality:** Single input field for long URL and a "**Shorten**" button .**Result:** Displays
- and copies shortened link

B. Monitoring Dashboard (Operator UX - Grafana)

The Grafana dashboard will provide an operator's view of service health and usage patterns, built from the Prometheus

:metrics

Insight Provided	Metric / Visualization	Panel Title
Shows traffic spikes	Rate of `url_creation_total` over 5 min	Usage Rate
Shows total usage	Single stat of latest `url_creation_total`	Total Links
Monitors user experience	Graph of **95th Percentile** for `request_latency_seconds`	P95 Latency
Key indicator for issues	Graph of `redirect_failure_total` rate	Service Health / Errors

Marketing and Project Visuals (Imagery Concepts) .2

Marketing Blurb

.Shortify: Build. Ship. Monitor. Total Control Over Your URLs

Tired of deploying web services into the dark? Shortify is the all-in-one solution that proves it can be simple. We deliver a .high-speed URL shortener service fully encapsulated in **Docker containers**

.But our true power is observability

:With **Prometheus** collecting vital signs and **Grafana** providing clear, beautiful dashboards, you get

- .**Real-Time Insight:** Instantly see usage trends and traffic peaks
- .**Performance Assurance:** Monitor 95th percentile latency to guarantee a fast experience
- .**Operational Confidence:** Immediate alerts for critical issues like rising 404 errors